网络建设与调试
报告单

示例：(仅供参考，具体以正式赛卷报告单为准)

在 SW1 上执行 show vlan 命令收集信息，截图如下：

```
SW1
文件(F)  编辑(E)  视图(V)  搜索(S)  终端(T)  标签(B)  帮助(H)
SW1                                   d526-2@d526-2-pc: ~/桌面
SW1(config)#
SW1(config)#show vlan
VLAN Name             Type        Media      Ports
---- ------------     ----------  ---------  ----------------------------------------
1    default          Static      ENET       Ethernet1/0/1       Ethernet1/0/2
                                             Ethernet1/0/3       Ethernet1/0/4
                                             Ethernet1/0/5       Ethernet1/0/6
                                             Ethernet1/0/7       Ethernet1/0/8
                                             Ethernet1/0/9       Ethernet1/0/10
                                             Ethernet1/0/11      Ethernet1/0/12
                                             Ethernet1/0/13      Ethernet1/0/14
                                             Ethernet1/0/15      Ethernet1/0/16
                                             Ethernet1/0/17      Ethernet1/0/18
                                             Ethernet1/0/19      Ethernet1/0/20
                                             Ethernet1/0/21      Ethernet1/0/22
                                             Ethernet1/0/23      Ethernet1/0/24
                                             Ethernet1/0/25      Ethernet1/0/26
                                             Ethernet1/0/27      Ethernet1/0/28
10   BJ-1             Static      ENET
20   BJ-2             Static      ENET
30   BJ-3             Static      ENET
40   BJ-4             Static      ENET
50   BJ-5             Static      ENET
SW1(config)#
SW1(config)#
```

- **请注意尽量保证格式、图片排列整齐**
- **请注意尽量窗口只显示需要查看的信息，窗口内的显示内容大小合适，清晰！**
- **为保证结果真实性，提供图片截图，不要复制命令行配置到本窗口，否则视为零分。**
- **为更好识别窗口当前对应设备，可以适当地给终端标题设置对应的名称**

一、交换配置

1、在 SW1 上执行 show vlan、show run | include multi 命令收集信息，截图如下：（10 分）

方框所有 10 个 vlan 和端口对应一致全部正确得 5 分，少一个 VLAN 或端口扣 0.5 分；
方框查询有开启 multi config access 命令正确得 5 分。

```
SW1#show vlan
VLAN Name             Type        Media     Ports
---- ------------     ---------   --------  ----------------------------------------
1    default          Static      ENET      Ethernet1/0/9        Ethernet1/0/10
                                            Ethernet1/0/11       Ethernet1/0/12
                                            Ethernet1/0/13       Ethernet1/0/14
                                            Ethernet1/0/15       Ethernet1/0/16
                                            Ethernet1/0/17       Ethernet1/0/18
                                            Ethernet1/0/21       Ethernet1/0/22
                                            Ethernet1/0/23       Ethernet1/0/24
                                            Ethernet1/0/25       Ethernet1/0/26
                                            Ethernet1/0/27       Ethernet1/0/28
10   VLAN0010         Static      ENET      Ethernet1/0/1        Ethernet1/0/9(T)
                                            Ethernet1/0/10(T)
20   VLAN0020         Static      ENET      Ethernet1/0/2        Ethernet1/0/9(T)
                                            Ethernet1/0/10(T)
30   VLAN0030         Static      ENET      Ethernet1/0/3        Ethernet1/0/9(T)
                                            Ethernet1/0/10(T)
40   VLAN0040         Static      ENET      Ethernet1/0/4        Ethernet1/0/9(T)
                                            Ethernet1/0/10(T)
50   VLAN0050         Static      ENET      Ethernet1/0/5        Ethernet1/0/9(T)
                                            Ethernet1/0/10(T)
60   VLAN0060         Static      ENET      Ethernet1/0/6        Ethernet1/0/9(T)
                                            Ethernet1/0/10(T)
70   VLAN0070         Static      ENET      Ethernet1/0/7        Ethernet1/0/9(T)
                                            Ethernet1/0/10(T)
80   VLAN0080         Static      ENET      Ethernet1/0/8        Ethernet1/0/9(T)
                                            Ethernet1/0/10(T)
1019 VLAN1019         Static      ENET      Ethernet1/0/19
1020 VLAN1020         Static      ENET      Ethernet1/0/20
SW1#show run | include mult
multi config access
SW1#
SW1#
```

2、在 SW3 上执行 show span mst config、show span mst 10、show span mst 22 命令收集信息，截图如下：（10 分）

方框内 Name 和 Rev 配置正确得 1 分、实例 10 和 22 规划 vlan 正确得 1 分。
方框内实例 10 的 root id 是 4096 且 10 端口为 FWD 状态，11 为 BLK 状态正确得 4 分。
方框内实例 22 的 root id 是 4096 且 10 端口为 BLK 状态，11 为 FWD 状态正确得 4 分。

```
SW3#show spanning-tree mst config

Name          2025
Revision      1
Instance      Vlans Mapped
----------------------------------
00            1-9, 11-19, 21-29, 31-39, 41-49, 51-59, 61-69, 71-79, 81-4094
10            10, 20, 30, 40
22            50, 60, 70, 80
----------------------------------
```

```
SW3#show spanning-tree mst 10
************************************ Process 0 ************************************
############################ Instance 10 ############################
vlans mapped     : 10;20;30;40
Self Bridge Id   : 32768.00:03:0f:91:d5:10
Root Id          : 4096.00:03:0f:2a:77:ed
   PortName            ID     IntRPC    State Role     DsgBridge          DsgPort
--------------- ------------ --------- --- ---- ------------------ -----------
 Ethernet1/0/10   128.00010         0 FWD  ROOT  4096.00030f2a77ed   128.00010
 Ethernet1/0/11   128.00011     20000 BLK  ALTR  8192.00030f2a8453   128.00011

SW3#show spanning-tree mst 22
************************************ Process 0 ************************************
############################ Instance 22 ############################
vlans mapped     : 50;60;70;80
Self Bridge Id   : 32768.00:03:0f:91:d5:10
Root Id          : 4096.00:03:0f:2a:84:53
   PortName            ID     IntRPC    State Role     DsgBridge          DsgPort
--------------- ------------ --------- --- ---- ------------------ -----------
 Ethernet1/0/10   128.00010     20000 BLK  ALTR  8192.00030f2a77ed   128.00010
 Ethernet1/0/11   128.00011         0 FWD  ROOT  4096.00030f2a8453   128.00011

SW3#
```

3-1、在 SW1 上执行 show vrrp 10、show vrrp 20、show vrrp 30、show vrrp 40 命令收集信息，截图如下：（8 分）

每个 VRRP 组正确得 2 分，每个 VRRP 组中参数错误不得分。

```
SW1#show vrrp 10
VrId 10
 State is Master
 Virtual IP is 10.255.11.253 (Not IP owner)
 Interface is Vlan10
 Configured priority is 254, Current priority is 254
 Advertisement interval is 1 sec
 Preempt mode is TRUE
 Circuit failover interface Vlan1019, Priority Delta 204, Status UP

SW1#show vrrp 20
VrId 20
 State is Master
 Virtual IP is 10.255.12.253 (Not IP owner)
 Interface is Vlan20
 Configured priority is 254, Current priority is 254
 Advertisement interval is 1 sec
 Preempt mode is TRUE
 Circuit failover interface Vlan1019, Priority Delta 204, Status UP
```

```
SW1#show vrrp 30
VrId 30
 State is Master
 Virtual IP is 10.255.13.253 (Not IP owner)
 Interface is Vlan30
 Configured priority is 254, Current priority is 254
 Advertisement interval is 1 sec
 Preempt mode is TRUE
 Circuit failover interface Vlan1019, Priority Delta 204, Status UP

SW1#show vrrp 40
VrId 40
 State is Master
 Virtual IP is 10.255.14.253 (Not IP owner)
 Interface is Vlan40
 Configured priority is 254, Current priority is 254
 Advertisement interval is 1 sec
 Preempt mode is TRUE
 Circuit failover interface Vlan1019, Priority Delta 204, Status UP
```

3-2、在 SW2 上执行 show vrrp 50、show vrrp 60、show vrrp 70、show vrrp 80 命令收集信息，截图如下：（8 分）

每个 VRRP 组正确得 2 分，每个 VRRP 组中参数错误不得分。

```
SW2#show vrrp 50
VrId 50
 State is Master
 Virtual IP is 10.255.15.253 (Not IP owner)
 Interface is Vlan50
 Configured priority is 254, Current priority is 254
 Advertisement interval is 1 sec
 Preempt mode is TRUE
 Circuit failover interface Vlan1019, Priority Delta 204, Status UP

SW2#show vrrp 60
VrId 60
 State is Master
 Virtual IP is 10.255.16.253 (Not IP owner)
 Interface is Vlan60
 Configured priority is 254, Current priority is 254
 Advertisement interval is 1 sec
 Preempt mode is TRUE
 Circuit failover interface Vlan1019, Priority Delta 204, Status UP
```

```
SW2#show vrrp 70
VrId 70
 State is Master
 Virtual IP is 10.255.17.253 (Not IP owner)
 Interface is Vlan70
 Configured priority is 254, Current priority is 254
 Advertisement interval is 1 sec
 Preempt mode is TRUE
 Circuit failover interface Vlan1019, Priority Delta 204, Status UP

SW2#show vrrp 80
VrId 80
 State is Master
 Virtual IP is 10.255.18.253 (Not IP owner)
 Interface is Vlan80
 Configured priority is 254, Current priority is 254
 Advertisement interval is 1 sec
 Preempt mode is TRUE
 Circuit failover interface Vlan1019, Priority Delta 204, Status UP
```

4、在 SW3 上执行 show ip route vrf Internet、show ip route vrf Storage、show ip route vrf RemotUser 命令收集信息，截图如下：（11 分）

三个实例当中总共 11 条对应的直连路由，正确得 11 分。

```
SW3#show ip route vrf Internet
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

C       200.200.3.3/32 is directly connected, Loopback3  tag:0
C       200.200.200.0/30 is directly connected, Vlan1017  tag:0
C       200.200.200.4/30 is directly connected, Vlan1018  tag:0
S       200.200.200.16/28 [1/0] via 200.200.200.2, Vlan1017  tag:0
Total routes are : 4 item(s)
```

```
SW3#show ip route vrf Storage
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS int
       * - candidate default

Gateway of last resort is 10.255.255.50 to network 0.0.0.0

O*E1    0.0.0.0/0 [110/4] via 10.255.255.50, Vlan1016, 00:19:00  tag:0
C       10.255.3.1/32 is directly connected, Loopback1  tag:0
C       10.255.255.48/30 is directly connected, Vlan1016  tag:0
C       172.16.210.0/24 is directly connected, Vlan210  tag:0
C       172.16.220.0/24 is directly connected, Vlan220  tag:0
Total routes are : 5 item(s)
```

```
SW3#show ip route vrf RemotUser
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

Gateway of last resort is 10.255.255.45 to network 0.0.0.0

O*E1    0.0.0.0/0 [110/102] via 10.255.255.45, Vlan1015, 00:19:49  tag:0
C       10.255.3.2/32 is directly connected, Loopback2  tag:0
O       10.255.7.1/32 [110/2] via 10.255.255.45, Vlan1015, 00:19:49  tag:0
O       10.255.255.24/30 [110/2] via 10.255.255.45, Vlan1015, 00:19:49  tag:0
C       10.255.255.44/30 is directly connected, Vlan1015  tag:0
C       172.16.130.0/24 is directly connected, Vlan130  tag:0
C       172.16.140.0/24 is directly connected, Vlan140  tag:0
Total routes are : 7 item(s)
```

5、在 SW1 上执行 show run | include snmp-server 命令收集信息，截图如下：（5 分）

总共 11 条对应的配置，全部正确得 5 分，加密的字符串不计。错一条或漏一条扣 0.4 分

```
SW1#show run | include snmp-server
snmp-server enable
snmp-server securityip 10.255.100.25
snmp-server securityip 2001:10:255:100::25
snmp-server trap-source 10.255.1.1
snmp-server trap-source 2001:10:255:1::1
snmp-server engineid 1001
snmp-server user U2025 G2025 authPriv 3des 7b24972a017861eada913627f5bf47be auth md5 7b24972
a017861eada913627f5bf47be
snmp-server group G2025 authpriv read S_R write S_W
snmp-server host 10.255.100.25 v3 authpriv U2025
snmp-server host 2001:10:255:100::25 v3 authpriv U2025
snmp-server enable traps
```

6、在 SW1 上执行 show run | include urpf 命令收集信息，截图如下：（8 分）

```
SW1#show run | include urpf
urpf enable
```

7、在 SW1 上执行 show run | include interrupt 命令收集信息，截图如下：（9 分）

```
SW1#show run | include interrupt
port-scan-mode interrupt
```

8、在 SW1 上执行 show access-lists 、show vacl vlan 命令收集信息，截图如下：（11 分）

框 1 全部正确得 6 分，少一条扣 1 分；框 2 全部正确得 5 分否则不得分；

```
SW1#show access-lists
ip access-list extended ruchong(used 8 time(s)) 6 rule(s)
   rule ID 1: deny tcp any-source any-destination d-port 445
   rule ID 2: deny tcp any-source any-destination d-port 135
   rule ID 3: deny tcp any-source any-destination d-port 139
   rule ID 4: deny udp any-source any-destination d-port 137
   rule ID 5: deny udp any-source any-destination d-port 138
   rule ID 6: permit ip any-source any-destination

SW1#show vacl vlan
VLAN 10:
IP Ingress access-list used is ruchong, traffic-statistics Disable.
IP Egress access-list used is ruchong,traffic-statistics Disable.

VLAN 20:
IP Ingress access-list used is ruchong, traffic-statistics Disable.
IP Egress access-list used is ruchong,traffic-statistics Disable.

VLAN 30:
IP Ingress access-list used is ruchong, traffic-statistics Disable.
IP Egress access-list used is ruchong,traffic-statistics Disable.

VLAN 40:
IP Ingress access-list used is ruchong, traffic-statistics Disable.
IP Egress access-list used is ruchong,traffic-statistics Disable.
```

二、路由配置（请完成整体网络架构路由调试再截图）

1-1、在 FW1 上执行 show config int e0/1 | include manage 命令收集信息，截图如下：（2 分）

查询端口管理 4 个服务正确得 2 分。

```
FW1# show conf int e0/1 | include manage

  manage ping
  manage ssh
  manage http
  manage https
```

1-2、在 SW1 上执行 show run | include ssh-server 命令收集信息，截图如下；（1 分）

查询 ssh 服务开启成功正确得 1 分

```
SW1#show run | include ssh-server
ssh-server enable
```

1-3、在 RT1 上执行 ssh -d 10.255.255.30 - admin -v 2 并成功登录到 RT2 全局配置模式的效果信息截图，截图如下：（2 分）

在 RT1 上 ssh 登录 RT2 成功正确进入，得 2 分。否则 0 分。

```
RT1#ssh -d 10.255.255.34 - admin -v 2
Connecting to remote host...
Press 'q' or 'Q' to quit connection.
enter your password:
            Welcome to Digital China Multi-Protocol DCR-2855 Series

RT2>ena
RT2#conf
RT2_config#
```

SSH地址改成10.255.255.30

2、在 RT1 上执行 show run | include snmp-server 命令收集信息，截图如下：（3 分）

在 RT2 上查询 6 条配置正确得 3 分。

```
snmp-server group G2025 v3 priv read S_R write S_W
snmp-server user U2025 G2025 v3 auth sha 0 Key-2025
snmp-server hostv6 2001:10:255:100::25 version v3 auth U2025
snmp-server host 10.255.100.25 version v3 auth U2025
snmp-server trap-source Loopback1
snmp-server engineID local 1000
```

3-1、在 SW1 上执行 show ipv6 inter bri 命令收集信息，截图如下：（5 分）

在 SW1 上查询 10 个接口的 ipv6 地址配置正确得 5 分，错一个扣 0.5 分。

```
SW1#show ipv6 inter bri
Ethernet0                              [down]
     unassigned
Vlan10                                 [up]
     2001:10:255:11::251/64
     fe80::203:fff:fe2a:77ec/64
Vlan20                                 [up]
     2001:10:255:12::251/64
     fe80::203:fff:fe2a:77ec/64
Vlan30                                 [up]
     2001:10:255:13::251/64
     fe80::203:fff:fe2a:77ec/64
Vlan40                                 [up]
     2001:10:255:14::251/64
     fe80::203:fff:fe2a:77ec/64
Vlan50                                 [up]
     2001:10:255:15::251/64
     fe80::203:fff:fe2a:77ec/64
Vlan60                                 [up]
     2001:10:255:16::251/64
     fe80::203:fff:fe2a:77ec/64
Vlan70                                 [up]
     2001:10:255:17::251/64
     fe80::203:fff:fe2a:77ec/64
Vlan80                                 [up]
     2001:10:255:18::251/64
     fe80::203:fff:fe2a:77ec/64
Vlan1019                               [up]
     fe80::203:fff:fe2a:77ec/64
Vlan1020                               [up]
     fe80::203:fff:fe2a:77ec/64
Loopback                               [up]
     ::1/128
Loopback1                              [up]
     2001:10:255:1::1/128
     fe80::203:fff:fe2a:77ec/64
Loopback2                              [up]
     2001:10:255:1::2/128
     fe80::203:fff:fe2a:77ec/64
```

3-2、在 SW2 上执行 show ipv6 inter bri 命令收集信息，截图如下：（5 分）

在 SW2 上查询 10 个接口的 ipv6 地址配置正确得 5 分，错一个扣 0.5 分。

```
SW2#show ipv6 inter bri
Ethernet0                              [down]
    unassigned
Vlan10                                 [up]
    2001:10:255:11::252/64
    fe80::203:fff:fe2a:8452/64
Vlan20                                 [up]
    2001:10:255:12::252/64
    fe80::203:fff:fe2a:8452/64
Vlan30                                 [up]
    2001:10:255:13::252/64
    fe80::203:fff:fe2a:8452/64
Vlan40                                 [up]
    2001:10:255:14::252/64
    fe80::203:fff:fe2a:8452/64
Vlan50                                 [up]
    2001:10:255:15::252/64
    fe80::203:fff:fe2a:8452/64
Vlan60                                 [up]
    2001:10:255:16::252/64
    fe80::203:fff:fe2a:8452/64
Vlan70                                 [up]
    2001:10:255:17::252/64
    fe80::203:fff:fe2a:8452/64
Vlan80                                 [up]
    2001:10:255:18::252/64
    fe80::203:fff:fe2a:8452/64
Vlan1019                               [up]
    fe80::203:fff:fe2a:8452/64
Vlan1020                               [up]
    fe80::203:fff:fe2a:8452/64
Loopback                               [up]
    ::1/128
Loopback1                              [up]
    2001:10:255:2::1/128
    fe80::203:fff:fe2a:8452/64
Loopback2                              [up]
    2001:10:255:2::2/128
    fe80::203:fff:fe2a:8452/64
```

4、将 PC1 接在 SW1 上的 E1/0/1 口，PC2 接在 SW2 上的 E1/0/7 口进行 IPv4 和 IPv6 地址的自动地址获取。在 SW2 上执行 show ip dhcp binding、show ipv6 dhcp binding 命令收集信息，截图如下：（4 分）

下图 1：在 SW2 查询 ipv6 终端获取地址，两个固定正确得 2 分

下图 2：在 SW2 查询 ipv4 终端获取地址，两个固定正确得 2 分

```
SW2#
SW2#show ipv6 dhcp binding

Creating DHCPv6 binding table...

The number of DHCPv6 bindings is 2
Client: iatype IANA, iaid 0x04f46b8c
  DUID: 00:01:00:01:2d:04:83:a9:f4:6b:8c:89:48:50
  IANA leased address: 2001:10:255:11::10
  Preferred lifetime 429496729 5 seconds, valid lifetime 4294967295 seconds
  Lease obtained at %Feb 02 15:59:13 2024
  Lease expires at %Feb 02 10:13:36 1924  (4294966851 seconds left)
Client: iatype IANA, iaid 0x249649d3
  DUID: 00:04:c0:cc:b5:8c:62:2a:13:2c:04:88:9b:72:79:c5:73:66
  IANA leased address: 2001:10:255:17::10
  Preferred lifetime 604800 seconds, valid lifetime 2592000 seconds
  Lease obtained at %Feb 02 16:02:04 2024
  Lease expires at %Mar 03 16:02:04 2024  (2591727 seconds left)
SW2#
SW2#
SW2#
```

```
SW2#
SW2#show ip dhcp binding
Total dhcp binding items: 2, the matched: 2
IP address          Hardware address        Lease expiration        Type
10.255.11.10        01-01-01-01-01-01       Infinite                Manual
10.255.17.10        E8-6A-64-76-F7-C4       Infinite                Manual
SW2#
SW2#
SW2#
SW2#
```

5-1、在 RT1 上执行 show ip ospf nei、show ipv6 ospf nei 命令收集信息，截图如下：（11 分）

下图 2：查询 ipv4 邻居建立为 Full 状态， 4 个邻居进程和区域所属建立正确得 8 分。

下图 1：查询 ipv:6 邻居建立为 Full 状态，3 个邻居进程和区域所属建立正确得 3 分。

```
RT1#show ipv6 ospf neighbor
OSPFv3 Process (1)
Neighbor ID     Pri   State           Dead Time   Interface        Instance ID
10.255.2.1        1   Full/Backup     00:00:34    GigaEthernet0/1  0
10.255.4.1        1   Full/Backup     00:00:32    GigaEthernet0/2  0
OSPFv3 Process (2)
Neighbor ID     Pri   State           Dead Time   Interface        Instance ID
10.255.7.1        1   Full/DR         00:00:38    GigaEthernet0/3  0
RT1#
```

```
RT1#show ip ospf neighbor
------------------------------------------------------------------------
                                OSPF process: 1

                                 AREA: 0
Neighbor ID      Pri   State            DeadTime   Neighbor Addr   Interface
10.255.2.1       1     FULL/BDR         39         10.255.255.22   GigaEthernet0/1
10.255.4.1       1     FULL/BDR         34         10.255.255.17   GigaEthernet0/2

                                 AREA: 1
Neighbor ID      Pri   State            DeadTime   Neighbor Addr   Interface
10.255.6.1       1     FULL/DR          32         10.255.255.30   GigaEthernet0/0
------------------------------------------------------------------------
                                OSPF process: 2

                                 AREA: 2
Neighbor ID      Pri   State            DeadTime   Neighbor Addr   Interface
10.255.7.1       1     FULL/DR          39         10.255.255.26   GigaEthernet0/3
------------------------------------------------------------------------
```

5-2、在 SW3 上执行 show run route ospf 命令收集信息，截图：（5 分）

查询进程下的网络宣告，两个所属区域 2 正确得 5 分。

```
SW3#show run route ospf
!
router ospf 1 Storage
 ospf router-id 10.255.3.1
 network 10.255.3.1/32 area 1
 network 10.255.255.48/30 area 1
 network 172.16.210.0/24 area 1
 network 172.16.220.0/24 area 1
 filter-policy route-map Default
!
router ospf 2 RemotUser
 ospf router-id 10.255.3.2
 network 10.255.3.2/32 area 2
 network 10.255.255.44/30 area 2
 network 172.16.130.0/24 area 2
 network 172.16.140.0/24 area 2
!
```

5-3、在 RT1 上执行 show ipv6 route ospf 命令收集信息，截图：（5 分）

查询 ipv6 路由，有 OE1 路由正确得 1 分，其他 4 条路由正确得 4 分。

```
RT1#show ipv6 route ospf
OE1     ::/0[1]
         [110,21] via fe80::203:fff:fe84:a552(on GigaEthernet0/2)
O       2001:10:255:1::1/128[2]
         [110,2] via fe80::203:fff:fe84:a552(on GigaEthernet0/2)
O       2001:10:255:2::1/128[2]
         [110,1] via fe80::203:fff:fe2a:8452(on GigaEthernet0/1)
O       2001:10:255:4::1/128[1]
         [110,2] via fe80::203:fff:fe84:a552(on GigaEthernet0/2)
O       2001:10:255:7::1/128[1]
         [110,2] via fe80::203:fff:fea4:f601(on GigaEthernet0/3)
RT1#
```

5-4、在 FW2 上执行 show ip route ospf；在 RT1 上执行 show ip prefix-list、show run | include redistribute 命令收集信息，截图如下：（8 分）

下图 1：查询 ospf 路由，正确得 4 分。

下图 2：查询方框内前缀名称和列表，两个前缀正确得 2 分，错一个扣 0.5 分。查询方框内有两条 redistribute 完全正确得 2 分。

```
FW2# show ip route ospf
Codes: K - kernel route, C - connected, S - static, Z - ISP, R - RIP, O - OSPF,
       B - BGP, D - DHCP, P - PPPoE, W - wireless, H - HOST, G - SCVPN, V - VPN
       I - ISIS, Y - SYNC, L - llb outbound, > - selected first nexthop, * - FI
BFD enable

Routing Table for Virtual Router <trust-vr>
================================================================================
O>* 0.0.0.0/0 [110/101/1] via 10.255.255.25, ethernet0/1, 00:57:34
O>* 10.255.3.2/32 [110/2/1] via 10.255.255.46, ethernet0/2, 00:57:35
O>* 172.16.130.0/24 [110/2/1] via 10.255.255.46, ethernet0/2, 00:57:35
O>* 172.16.140.0/24 [110/2/1] via 10.255.255.46, ethernet0/2, 00:57:35
================================================================================
```

```
RT1#show ip prefix-list
ip prefix-list RT1-FW2-IPv4: 1 entries
   seq 5 permit 10.255.255.24/30
ip prefix-list SW2-RT1-IPv4: 4 entries
   seq 5 permit 10.255.15.0/24
   seq 10 permit 10.255.16.0/24
   seq 15 permit 10.255.18.0/24
   seq 20 permit 10.255.17.0/24
ip prefix-list SW3-FW2-IPv4: 3 entries
   seq 5 permit 10.255.7.1/32
   seq 10 permit 172.16.130.0/24
   seq 15 permit 172.16.140.0/24
RT1#show run | include redistribute
Building configuration...

Current configuration:
!
 redistribute ospf 2 route-map SW3-FW2-IPv4
 redistribute connect route-map RT1-FW2-IPv4
 redistribute ospf 2
```

5-5、在 FW2 上执行 show ipv6 route ospf、命令收集信息，截图如下：（1 分）

下图 1：查询 ospf ipv6 路由，正确得 1 分。

```
FW2# show ipv6 route ospf
Codes: K - kernel route, C - connected, S - static, R - RIPng, O - OSPFv3,
       B - BGP, I - ISIS, A - AUTOCONF, H - HOST, > - selected route,
       * - FIB route,

Routing Table for Virtual Router <trust-vr>
================================================================================
O>   ::/0
  * via FE80::203:FFF:FEE0:ECAC, ethernet0/1 [110/151/1]
================================================================================
```

5-6、在 SW1 上执行 show run int vlan 1020 、show ip route ospf 命令收集信息，截图如下：（6 分）

Vlan1020 接口下配置正确得 1 分。路由表共 22 条，下一跳全部正确得 5 分。错 1 条扣 0.5 分，扣完即止。

```
SW1#show run int vlan 1020
!
interface Vlan1020
 ip ospf cost 100
 ipv6 ospf cost 100
 ipv6 router ospf area 0 tag 1
 ip address 10.255.255.1 255.255.255.252
!
SW1#show ip route ospf
O*E1    0.0.0.0/0 [110/2] via 10.255.255.13, Vlan1019, 01:04:45  tag:0
O       10.255.2.1/32 [110/4] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O IA    10.255.3.1/32 [110/5] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O       10.255.4.1/32 [110/2] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O       10.255.5.1/32 [110/3] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O IA    10.255.6.1/32 [110/4] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O E2    10.255.7.1/32 [110/100] via 10.255.255.13, Vlan1019, 01:04:45  tag:0
O IA    10.255.8.1/32 [110/5] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O       10.255.255.16/30 [110/2] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O       10.255.255.20/30 [110/3] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O E2    10.255.255.24/30 [110/100] via 10.255.255.13, Vlan1019, 01:04:45  tag:0
O IA    10.255.255.28/30 [110/3] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O IA    10.255.255.40/30 [110/4] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O IA    10.255.255.48/30 [110/4] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O E2    172.16.130.0/24 [110/100] via 10.255.255.13, Vlan1019, 01:04:42  tag:0
O E2    172.16.140.0/24 [110/100] via 10.255.255.13, Vlan1019, 01:04:42  tag:0
O IA    172.16.150.0/24 [110/5] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O IA    172.16.160.0/24 [110/5] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O IA    172.16.170.0/24 [110/5] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O IA    172.16.180.0/24 [110/5] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O IA    172.16.210.0/24 [110/5] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
O IA    172.16.220.0/24 [110/5] via 10.255.255.13, Vlan1019, 01:04:46  tag:0
Total routes are : 22 item(s)
SW1#
```

5-7、在 SW2 上执行 show run int vlan 1019、show run int vlan 1020、show ip route ospf 命令收集信息，截图如下：（8 分）

Vlan1019 和 Vlan1020 接口下配置正确得 2 分，路由表共 26 条，下一跳全部正确得 6 分。错 1 条扣 0.5 分，扣完即止。

```
SW2#show run int vlan 1019
!
interface Vlan1019
 ip ospf cost 99
 ipv6 ospf cost 99
 ipv6 router ospf area 0 tag 1
 ip address 10.255.255.22 255.255.255.252
!
SW2#show run int vlan 1020
!
interface Vlan1020
 ip ospf cost 100
 ipv6 ospf cost 100
 ipv6 router ospf area 0 tag 1
 ip address 10.255.255.2 255.255.255.252
!
```

```
SW2#show ip route ospf
O*E1    0.0.0.0/0 [110/101] via 10.255.255.21, Vlan1019, 01:41:25  tag:0
O       10.255.1.1/32 [110/101] via 10.255.255.1, Vlan1020, 01:41:56  tag:0
O IA    10.255.3.1/32 [110/102] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O       10.255.4.1/32 [110/101] via 10.255.255.21, Vlan1019, 01:41:26  tag:0
O       10.255.5.1/32 [110/100] via 10.255.255.21, Vlan1019, 01:41:56  tag:0
O IA    10.255.6.1/32 [110/101] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O E2    10.255.7.1/32 [110/100] via 10.255.255.21, Vlan1019, 01:41:06  tag:0
O IA    10.255.8.1/32 [110/102] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O       10.255.11.253/32 [110/101] via 10.255.255.1, Vlan1020, 01:41:56  tag:0
O       10.255.12.253/32 [110/101] via 10.255.255.1, Vlan1020, 01:41:56  tag:0
O       10.255.13.253/32 [110/101] via 10.255.255.1, Vlan1020, 01:41:56  tag:0
O       10.255.14.253/32 [110/101] via 10.255.255.1, Vlan1020, 01:41:56  tag:0
O       10.255.255.12/30 [110/101] via 10.255.255.1, Vlan1020, 01:41:26  tag:0
                         [110/101] via 10.255.255.21, Vlan1019, 01:41:26  tag:0
O       10.255.255.16/30 [110/100] via 10.255.255.21, Vlan1019, 01:41:56  tag:0
O E2    10.255.255.24/30 [110/100] via 10.255.255.21, Vlan1019, 01:41:56  tag:0
O IA    10.255.255.28/30 [110/100] via 10.255.255.21, Vlan1019, 01:41:56  tag:0
O IA    10.255.255.40/30 [110/101] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O IA    10.255.255.48/30 [110/101] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O E2    172.16.130.0/24 [110/100] via 10.255.255.21, Vlan1019, 01:41:06  tag:0
O E2    172.16.140.0/24 [110/100] via 10.255.255.21, Vlan1019, 01:41:06  tag:0
O IA    172.16.150.0/24 [110/102] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O IA    172.16.160.0/24 [110/102] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O IA    172.16.170.0/24 [110/102] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O IA    172.16.180.0/24 [110/102] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O IA    172.16.210.0/24 [110/102] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
O IA    172.16.220.0/24 [110/102] via 10.255.255.21, Vlan1019, 01:41:51  tag:0
Total routes are : 27 item(s)
```

5-8、在 SW3 上执行 show ip route vrf Storage、show run | include Default 命令收集信息，截图如下：　（6 分）

```
SW3#
SW3#
SW3#
SW3#show ip route  vrf Storage
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

Gateway of last resort is 10.255.255.50 to network 0.0.0.0

O*E1    0.0.0.0/0 [110/4] via 10.255.255.50, Vlan1016, 00:52:24  tag:0
C       10.255.3.1/32 is directly connected, Loopback1  tag:0
C       10.255.9.1/32 is directly connected, Loopback11  tag:0
C       10.255.255.48/30 is directly connected, Vlan1016  tag:0
C       172.16.210.0/24 is directly connected, Vlan210  tag:0
C       172.16.220.0/24 is directly connected, Vlan220  tag:0
Total routes are : 6 item(s)
SW3#
SW3#
SW3#
SW3#
SW3#show run | include  Default
 filter-policy route-map Default
ip prefix-list Default seq 5 permit 0.0.0.0/0
ip prefix-list Default seq 10 deny any
route-map Default permit 10
 match ip address prefix-list Default
SW3#
SW3#
SW3#
```

只显示一条OE1默认路由，无其他ospf路由正确得4分。

前缀和Route-map配置正确得2分。

6-1、在 AC1 上执行 ping src 172.16.150.254 200.200.200.5、 traceroute 200.200.200.5 命令收集信息，截图如下：（3 分）

使用命令 Ping 测试成功得 2 分，路由追踪下一跳地址正确得 1 分。

```
AC1#ping src 172.16.150.254 200.200.200.5
Type ^c to abort.
Sending 5 56-byte ICMP Echos to 200.200.200.5, using source address 172.16.150.254, timeout
is 2 seconds.
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
AC1#traceroute 200.200.200.5
Type ^c to abort.
Traceroute to host 200.200.200.5, maxhops is 30, timeout is 2000ms.
 1   0ms    10.255.255.41
 2   0ms    200.200.200.5
Traceroute completed.
```

6-2、在 AC1 上执行 show run int vlan 100、show run int l1 命令收集信息，截图如下：（正确得 2 分）

```
AC1#show run int vlan 100
!
interface Vlan100
 ipv6 address 2001:172:16:100::254/64
 no ipv6 nd suppress-ra
 ipv6 nd managed-config-flag
 ipv6 nd other-config-flag
 ip address 172.16.100.254 255.255.255.0
 ipv6 dhcp server APv6
!
AC1#show run int l1
!
interface Loopback1
 ipv6 address 2001:10:255:8::1/128
 ip address 10.255.8.1 255.255.255.255
!
AC1#
```

7-1、在 R1 上执行 show ip bgp summary、show ip bgp ipv6 unicast summary 命令收集信息，截图如下：（7 分）

查询 bgp 邻居。ipv4（4 个）和 ipv6（3 个）对应的邻居地址和 AS 号正确得 7 分

```
RT1#show ip bgp summary
router bgp 65001
  BGP local AS: 65001, router identifier: 10.255.5.1
  IGP synchronization check: disabled
  Default local preference: 100
  Default maximum routes limit: 1280000
  Distance: external 20 internal 200 local 200
  Timers: keepalive 30, holdtime 90
  12 network entries using 3360 bytes of memory
  24 path entries using 1920 bytes of memory
  5 BGP path attribute entries using 1000 bytes of memory
  1 BGP AS-PATH entries using 24 bytes of memory
  0 BGP extended community entries using 0 bytes of memory

Neighbor        Ver          AS  State/Pref  Up/Down  FD  OutQ MsgRcvd MsgSent   VRF
10.255.1.1       4       65001           4 01:49:49   37     0     257     222
10.255.2.1       4       65001           4 01:49:52   36     0     257     222
10.255.4.1       4       65001           0 01:49:21   39     0     220     221
10.255.6.1       4       65002           0 01:49:24   38     0     219     222
```

```
RT1#show ip bgp ipv6 unicast summary
router bgp 65001
  BGP local AS: 65001, router identifier: 10.255.5.1
  IGP synchronization check: disabled
  Default local preference: 100
  Default maximum routes limit: 1280000
  Distance: external 20 internal 200 local 200
  Timers: keepalive 30, holdtime 90
  12 network entries using 3360 bytes of memory
  24 path entries using 1920 bytes of memory
  5 BGP path attribute entries using 1000 bytes of memory
  1 BGP AS-PATH entries using 24 bytes of memory
  0 BGP extended community entries using 0 bytes of memory

Neighbor        Ver          AS  State/Pref  Up/Down  FD  OutQ MsgRcvd MsgSent   VRF
2001:10:255:1::1
                 4       65001           8 01:59:39    34     0     281     242
2001:10:255:2::1
                 4       65001           8 01:59:39    35     0     281     242
2001:10:255:4::1
                 4       65001           0 01:58:47    40     0     239     240

  Number of configured neighbors: 3
```

7-2、在 RT1 上执行 show ipv6 route bgp 命令收集信息，截图：（4 分）

查询 bgp ipv6 路由，8 条全部正确得 4 分，少一条扣 0.5 分。

```
RT2#show ipv6 route bgp
B       2001:10:255:11::/64[1]
         [200,0] via fe80::203:fff:fe2a:8452(on GigaEthernet0/1)
         gate 2001:10:255:2::1
B       2001:10:255:12::/64[1]
         [200,0] via fe80::203:fff:fe2a:8452(on GigaEthernet0/1)
         gate 2001:10:255:2::1
B       2001:10:255:13::/64[1]
         [200,0] via fe80::203:fff:fe2a:8452(on GigaEthernet0/1)
         gate 2001:10:255:2::1
B       2001:10:255:14::/64[1]
         [200,0] via fe80::203:fff:fe2a:8452(on GigaEthernet0/1)
         gate 2001:10:255:2::1
B       2001:10:255:15::/64[1]
         [200,0] via fe80::203:fff:fe2a:8452(on GigaEthernet0/1)
         gate 2001:10:255:2::1
B       2001:10:255:16::/64[1]
         [200,0] via fe80::203:fff:fe2a:8452(on GigaEthernet0/1)
         gate 2001:10:255:2::1
B       2001:10:255:17::/64[1]
         [200,0] via fe80::203:fff:fe2a:8452(on GigaEthernet0/1)
         gate 2001:10:255:2::1
B       2001:10:255:18::/64[1]
         [200,0] via fe80::203:fff:fe2a:8452(on GigaEthernet0/1)
         gate 2001:10:255:2::1
```

7-3、在 RT1 和 FW 1 上各执行 show ip bgp、show ip route bgp 命令收集信息，截图如下：（6）

下图 1：查询 RT1 的 bgp 属性正确得 2 分，无属性不得分，且 bgp 路由表下一跳地址全部为 2.1 正确得 1 分。

下图 2：查询 FW1 的 bgp 属性正确得 2 分，无属性不得分，且 bgp 路由表下一跳地址全部为 2.1 正确得 1 分。

```
RT1#show ip bgp
BGP table version is 0, local router ID is 10.255.5.1
Status codes: s suppressed, d damped, h history, * valid, > best,
i - internal, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*>i10.255.15.0/24   10.255.2.1             100    100      0 i
* i                 10.255.1.1             200    100      0 i
*>i10.255.16.0/24   10.255.2.1             100    100      0 i
* i                 10.255.1.1             200    100      0 i
*>i10.255.17.0/24   10.255.2.1             100    100      0 i
* i                 10.255.1.1             200    100      0 i
*>i10.255.18.0/24   10.255.2.1             100    100      0 i
* i                 10.255.1.1             200    100      0 i

Total number of prefixes 4
RT1#show ip route bgp
VRF ID: 0

B       10.255.15.0/24        [200,100] via 10.255.2.1
B       10.255.16.0/24        [200,100] via 10.255.2.1
B       10.255.17.0/24        [200,100] via 10.255.2.1
B       10.255.18.0/24        [200,100] via 10.255.2.1
RT1#
```

路由对应的下一跳以及Metrice属性一致，得分否则不得分

```
FW1# show ip bgp
BGP Info, local router ID is 10.255.4.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

    Network          Next Hop            Metric LocPrf Weight Path

The total number of prefixes is 8
*>i 10.255.15.0/24   10.255.2.1             100    100      0 i
* i 10.255.15.0/24   10.255.1.1             200    100      0 i
*>i 10.255.16.0/24   10.255.2.1             100    100      0 i
* i 10.255.16.0/24   10.255.1.1             200    100      0 i
*>i 10.255.17.0/24   10.255.2.1             100    100      0 i
* i 10.255.17.0/24   10.255.1.1             200    100      0 i
*>i 10.255.18.0/24   10.255.2.1             100    100      0 i
* i 10.255.18.0/24   10.255.1.1             200    100      0 i
FW1# show ip route bgp
Codes: K - kernel route, C - connected, S - static, Z - ISP, R - RIP, O - OSPF,
       B - BGP, D - DHCP, P - PPPoE, W - wireless, H - HOST, G - SCVPN, V - VPN,
       I - ISIS, Y - SYNC, L - llb outbound, > - selected first nexthop, * - FIB
BFD enable

Routing Table for Virtual Router <trust-vr>
=============================================================================
B>* 10.255.15.0/24 [200/100/1] via 10.255.2.1
    (rgate : 10.255.255.18), 01:41:54
B>* 10.255.16.0/24 [200/100/1] via 10.255.2.1
    (rgate : 10.255.255.18), 01:41:54
B>* 10.255.17.0/24 [200/100/1] via 10.255.2.1
    (rgate : 10.255.255.18), 01:41:54
B>* 10.255.18.0/24 [200/100/1] via 10.255.2.1
    (rgate : 10.255.255.18), 01:41:54
=============================================================================
```

8、在 RT2 上各执行 show run int g0/3 命令收集信息，截图如下：（6 分）

```
RT2#show run in g0/3
Building configuration...

Current configuration:
!
interface GigaEthernet0/3
 ip address 200.200.200.6 255.255.255.252
 ipv6 enable
 ip accounting output-packets
 ip accounting access-violation
 ip accounting mac-address input
 ip accounting mac-address output
 ip nat outside
RT2#
```

三、无线配置（50 分）

1-1、在 AC1 上执行 show wireless、show wireless ap status 命令收集信息，截图如下：（10 分）

显示 4 个地址正确得 2 分，有 AP 管理成功状态 success 和 ip 地址正确得 8 分。

```
AC1#show wireless

Administrative Mode............................ Enable
Operational Status............................. Enabled
WS IP Address.................................. 10.255.8.1
WS IPv6 Address................................ 2001:10:255:8::1
WS Auto IP Assign Mode ........................ Enable
WS Switch Static IP ........................... 10.255.8.1
WS Switch Static IPv6 ......................... 2001:10:255:8::1
AP Authentication Mode......................... Mac
AP Auto Upgrade Mode........................... Disable
AP Independent Auto Upgrade Mode............... Disable
AP Validation Method........................... Local
Client Roam Timeout (secs)..................... 30
Country Code................................... CN - China
Peer Group ID.................................. 1
Cluster Priority............................... 1
Cluster Controller............................. Yes
Cluster Controller IP Address.................. 10.255.8.1
Cluster Controller IPv6 Address................ 2001:10:255:8::1
Wireless System IP control port................ 57775
Wireless Management Protocol................... TLS
AP Client QoS Mode............................. Disable
AP Igmp Snooping Mode.......................... Disable

AC1#show wireless ap status

    MAC Address                                                    Configuration

 (*) Peer Managed  IP Address                            Profile Status      Status
   Age
------------------ ------------------------------------- ------- ------- ------------- ---
-----------
 00-03-0f-8b-43-50 172.16.100.100                        1       Managed Success       0d:
00:00:00

Total Access Points............................ 1
```

1-2、在 AC1 全局模式上执行 ip dhcp pool team1 进入地址池模式下执行 show run curr 命令收集信息，截图如下：（8 分）

开启 service dhcp 服务正确得 2 分，6 个地址池名称和参数配置正确得 6 分。

```
AC1#conf
AC1(config)#ip dhcp pool team1
AC1(dhcp-team1-config)#show run curr
!
service dhcp
!
ip dhcp pool APv4
 network-address 172.16.100.0 255.255.255.0
 lease 7 0 0
 default-router 172.16.100.254
 dns-server 114.114.114.114
!
ip dhcp pool team1
 network-address 172.16.150.0 255.255.255.0
 lease 7 0 0
 default-router 172.16.150.254
 dns-server 114.114.114.114
!
ip dhcp pool team2
 network-address 172.16.160.0 255.255.255.0
 lease 7 0 0
 default-router 172.16.160.254
 dns-server 114.114.114.114
!
ip dhcp pool team3
 network-address 172.16.170.0 255.255.255.0
 lease 7 0 0
 default-router 172.16.170.254
 dns-server 114.114.114.114
!
ip dhcp pool Guest_int
 network-address 172.16.180.0 255.255.255.0
[redacted]
 default-router 172.16.180.254
 dns-server 114.114.114.114
!
ip dhcp pool AP1_IPv4
 host 172.16.100.100 255.255.255.0
 hardware-address 00-03-0F-8B-43-50
[redacted]
 default-router 172.16.100.254
 dns-server 114.114.114.114
!
```

1-3、在 AC1 上进入 ap profile 1 模式后执行 show run currnet 命令收集信息，在 AC1 上执行 show wireless ssid-database 命令收集信息，截图如下：（22 分）

7 个方框正确得 14 分，错一个框扣 2 分。4 个 ssid 配置正确 8 分。

```
AC1(config)#wireless
AC1(config-wireless)#ap profile 1
AC1(config-ap-profile)#show run curr
!
 ap profile 1
  name Default
  radio 1
   dot11n channel-bandwidth 20
   vap 0
    network 150
!
   vap 1
    enable
    network 160
!
   vap 2
    enable
    network 170
!
   vap 3
    enable
    network 180
!
!
  radio 2
   dot11n channel-bandwidth 40
   vap 0
    network 150
!
   vap 1
    enable
    network 160
!
   vap 2
    enable
    network 170
!
!
AC1(config-ap-profile)#
```

```
AC1(config-ap-profile)# show wireless ssid-database

SSID                            Action          status
------------------------------- --------------- ------------
Guest Network                   grant           managed
Managed SSID 2                  grant           managed
Managed SSID 3                  grant           managed
Managed SSID 4                  grant           managed
Managed SSID 5                  grant           managed
Managed SSID 6                  grant           managed
Managed SSID 7                  grant           managed
Managed SSID 8                  grant           managed
Managed SSID 9                  grant           managed
Managed SSID 10                 grant           managed
Managed SSID 11                 grant           managed
Managed SSID 12                 grant           managed
Managed SSID 13                 grant           managed
Managed SSID 14                 grant           managed
Managed SSID 15                 grant           managed
Managed SSID 16                 grant           managed
team1                           grant           managed
team2                           grant           managed
team3                           grant           managed
Guest_int                       grant           managed
AC1(config-ap-profile)#
```

2、在 AC1 上进入 wireless 模式后执行 show run currnet 命令收集信息，截图如下：（10 分）

Wireless 模式下参数配置正确得 10 分，enable 指令不纳入考核。

```
AC1(config-wireless)#show run curr
!
wireless
 enable
 discovery vlan-list 100
 static-ip  10.255.8.1
 static-ipv6  2001:10:255:8::1
 wireless ap anti-flood interval 15
 wireless ap anti-flood max-conn-count 10
 wireless ap anti-flood agetime 20
 network 1
!
```

四、安全配置（50 分）

1、在 FW1 上执行 show interface 、show snat 命令收集信息，截图如下：（10 分）

下图 1：查看接口列表 6 个 ip 地址和区域正确得 6 分。

下图 2：snatid 为 1 ，mode 模式为 DP-stky。各参数配置正确得 4 分。

```
FW1# show interface

H:physical state;A:admin state;L:link state;P:protocol state;U:up;D:down;K:ha keep up
==================================================================================================
Interface name        IP address/mask    Zone name        H A L P MAC address      Description
--------------------------------------------------------------------------------------------------
ethernet0/0           192.168.1.1/24     trust            U U U U 0003.0f84.a550   ------
ethernet0/1           10.255.255.13/30   trust            U U U U 0003.0f84.a551   ------
ethernet0/2           10.255.255.17/30   trust            U U U U 0003.0f84.a552   ------
ethernet0/3           200.200.200.2/30   untrust          U U U U 0003.0f84.a553   ------
ethernet0/4           0.0.0.0/0          NULL             D U D D 0003.0f84.a554   ------
ethernet0/5           0.0.0.0/0          NULL             D U D D 0003.0f84.a555   ------
ethernet0/6           0.0.0.0/0          NULL             D U D D 0003.0f84.a556   ------
ethernet0/7           0.0.0.0/0          NULL             D U D D 0003.0f84.a557   ------
ethernet0/8           0.0.0.0/0          NULL             D U D D 0003.0f84.a558   ------
loopback1             10.255.4.1/32      trust            U U U U --------------   ------
loopback2             10.255.4.2/32      trust            U U U U --------------   ------
tunnel1               172.16.99.254/24   VPNHub           U U U U --------------   ------
vswitchif1            0.0.0.0/0          NULL             D U D D 0003.0f84.a561   ------
==================================================================================================
FW1# show snat
--------------------------------------------------------------------------------------------------------------------------
vr name trust-vr
snat rules total number is :1
==========================================================================================================================
  Id from            to              service          egress if/vr     translate to     mode        start end   size
--------------------------------------------------------------------------------------------------------------------------
   1 0.0.0.0/0       0.0.0.0/0       Any              ethernet0/3      200.200.200.10~  DP-Stky
==========================================================================================================================
```

2、在 FW1 上执行 show policy 命令收集信息，截图如下：（10 分）

4 条策略各 2.5 分，共 10 分。每个策略源、目的区域、源地址、目的地址、服务正确得分。

```
FW1# show policy
Total rules count: 4
S: Rule Status (E - Enabled;  D - Disabled)
Flag: * - Need Application Identification
      S - Log Session Start;  E - Log Session End;  D - Log Policy Deny
      F - Drop Fragment;  P - Permit Unknown Application;  W - Web Redirect
Default action DENY. Default log OFF. Check to-self OFF. Session rematch ON
==========================================================================================================
S    Id Name            RBNS_Attr   Source          Destination     Service         Application  Action    Flag
----------------------------------------------------------------------------------------------------------
Any => Any
E    1                              0.0.0.0/0       0.0.0.0/0       Any                          PERMIT    ------
E    2                              ::/0            ::/0                                         PERMIT    ------
trust => untrust
E    3                              0.0.0.0/0       0.0.0.0/0       Any                          PERMIT    ------
VPNHub => trust
E    4                              0.0.0.0/0       10.255.11.0/24  Any                          PERMIT    ------
==========================================================================================================
```

3、在 FW2 上执行 show policy 命令收集信息，截图如下：（10 分）

2 条策略各 5 分，共 10 分。每个策略源、目的区域、源地址、目的地址、服务正确得分。

```
FW2# show policy
Total rules count: 2
S: Rule Status (E - Enabled;  D - Disabled)
Flag: * - Need Application Identification
      S - Log Session Start;  E - Log Session End;  D - Log Policy Deny
      F - Drop Fragment;  P - Permit Unknown Application;  W - Web Redirect
Default action DENY. Default log OFF. Check to-self OFF. Session rematch ON
==========================================================================================================
S    Id Name            RBNS_Attr   Source          Destination     Service         Application  Action    Flag
----------------------------------------------------------------------------------------------------------
trust => dmz
E    1                              0.0.0.0/0       0.0.0.0/0       Any                          PERMIT    ------
dmz => trust
E    2                              0.0.0.0/0       0.0.0.0/0       Any                          PERMIT    ------
==========================================================================================================
```

4、在 FW1 上执行 show track Track1、show config int e0/3 命令收集信息，截图如下：（10 分）

Show track Trakc1 查看 ip 地址、接口 I 置位为 5，T 置位为 10，4 项正确得 8 分，查看 3 接口下有 shutdown 关联 track 配置正确得 2 分。

```
FW1# show track Track1
==========================================================================================
Track name:Track1;track ID:2;local:no;threshold:255;delay threshold:255; bandwidth threshold:255;if ID:13;snat cnt:0
used type:shutdown; status:SUCCESSFUL; link_status: NORMAL;
I:interval;T:threshold;W:weight;S:status;F:failed;SU:successful;UN:unknown
M:mode;H:http;P:ping;A:arp;D:dns;T:tcp
HWMK:high watermark; LWMK:low watermark;DW:delay weight
FLAG:link status flag;N:normal;L:long-delay;O:overload

track ip:
------------------------------------------------------------------------------------------
----
M Track host      Track IP        PORT   OUT_IF          I   T   W   S   SRC_IF          PRIUSED_SRCIP   HWMK     LWM
------------------------------------------------------------------------------------------
----
P 200.200.200.1   200.200.200.1          ethernet0/3     5   10  255 SU
------------------------------------------------------------------------------------------
----
------------------------------------------------------------------------------------------
FW1# show conf int e0/3

interface ethernet0/3
  zone  "untrust"
  ip address 200.200.200.2 255.255.255.252
  shutdown track "Track1"
  ipv6 enable
  manage ping
  manage http
  manage https
  manage ssh
exit
```

5、在 FW1 上执行 show config | include u2025、show scvpn pool、show config int tun1 命令收集信息，截图如下：（10 分）

```
FW1# show config | include u2025

Building configuration..
Running configuration:
  user "u2025"
FW1# show scvpn pool
==============================================================================
Name                            Address range                   Netmask
------------------------------------------------------------------------------
sslpool                         172.16.99.100-172.16.99.200     255.255.255.0
==============================================================================
FW1# show conf int tun1

interface tunnel1
  zone  "VPNHub"
  ip address 172.16.99.254 255.255.255.0
  ipv6 enable
  manage ping
  tunnel scvpn "ssl2025"
exit
FW1#
```